

“CAREERCONNECT: JOB PORTAL MANAGEMENT SYSTEM”

School of Computing

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in
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By**

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May, 2026

BONAFIDE CERTIFICATE

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ABSTRACT

This project presents the design and development of a web-based Job Portal Application using Spring Boot, Thymeleaf, and MySQL. The system provides a platform for job seekers to search and apply for jobs, and for employers to post job openings and manage applications.

The application follows the Model-View-Controller (MVC) architecture, ensuring proper separation between the user interface, business logic, and data management. Spring Boot is used for backend development, Thymeleaf for dynamic web pages, and MySQL for database storage.

The system includes features such as user registration, login authentication, job posting, and application management. It improves the efficiency of recruitment processes and reduces manual effort.

Overall, the project demonstrates the effective use of modern web technologies in building a simple and efficient job portal system.

Keywords : Job Portal, Spring Boot, Thymeleaf, MySQL, MVC Architecture, Web Application

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LIST OF ACRONYMS AND ABBREVIATIONS

MVC	Model View Controller
API	Application Programming Interface
JPA	Java Persistence API
JDBC	Java Database Connectivity
SQL	Structured Query Language
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
HTTP	HyperText Transfer Protocol
DBMS	Database Management System

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Chapter 1

INTRODUCTION

1.1 Introduction

The rapid growth of the internet has significantly transformed the way recruitment and job searching are carried out. Traditional methods of job application and hiring are often time-consuming, inefficient, and lack proper organization. To overcome these challenges, a web-based Job Portal Application has been developed using modern technologies such as Spring Boot, Thymeleaf, and MySQL.

This application provides a centralized platform where job seekers can search and apply for jobs, while employers can post job vacancies and manage applications effectively. The system ensures seamless communication between users and improves the overall recruitment process. By automating various tasks, the application reduces manual effort and enhances efficiency.

The project follows the Model-View-Controller (MVC) architecture, which ensures proper separation of concerns between different layers of the application. The frontend is designed using Thymeleaf templates, the backend logic is implemented using Spring Boot, and MySQL is used for storing and managing data.

1.2 Aim of the Project

The main aim of this project is to design and develop a user-friendly and efficient Job Portal system that simplifies the recruitment process for both job seekers and employers.

The objectives of the project include:

- To provide a platform for users to register and log in securely.
- To enable job seekers to search and apply for jobs.
- To allow employers to post job vacancies and manage applicants.
- To maintain user and job data efficiently using a database system.
- To implement a scalable and maintainable web application using Spring Boot.

1.3 Scope of the Project

The scope of this project includes the development of a web-based application that facilitates job searching and recruitment activities. The system is designed to handle multiple users simultaneously and provide a responsive and interactive user experience.

The application covers:

- User registration and authentication.
- Job posting and management by employers.
- Job search and application by job seekers.
- Database management using MySQL.
- Basic security features for user data protection.

However, the system currently focuses on core functionalities and can be extended in the future with advanced features such as email notifications, resume uploads, and AI-based job recommendations.

1.4 Framework

The project is developed using the Spring Boot framework, which simplifies the development of Java-based web applications. It follows the MVC (Model-View-Controller) architecture:

- **Model:** Represents the data layer, including entities and database interactions.
- **View:** Developed using Thymeleaf templates to provide a dynamic user interface.
- **Controller:** Handles user requests and controls the application flow.

Spring Boot provides features such as dependency injection, embedded servers, and easy integration with databases, making it suitable for building scalable and efficient applications.

The overall system architecture of the Job Portal Application is shown in [Figure 1.1]. It illustrates the interaction between the user interface, controller, and database layers, ensuring efficient data flow within the system.

SYSTEM ARCHITECTURE OF JOB PORTAL APPLICATION

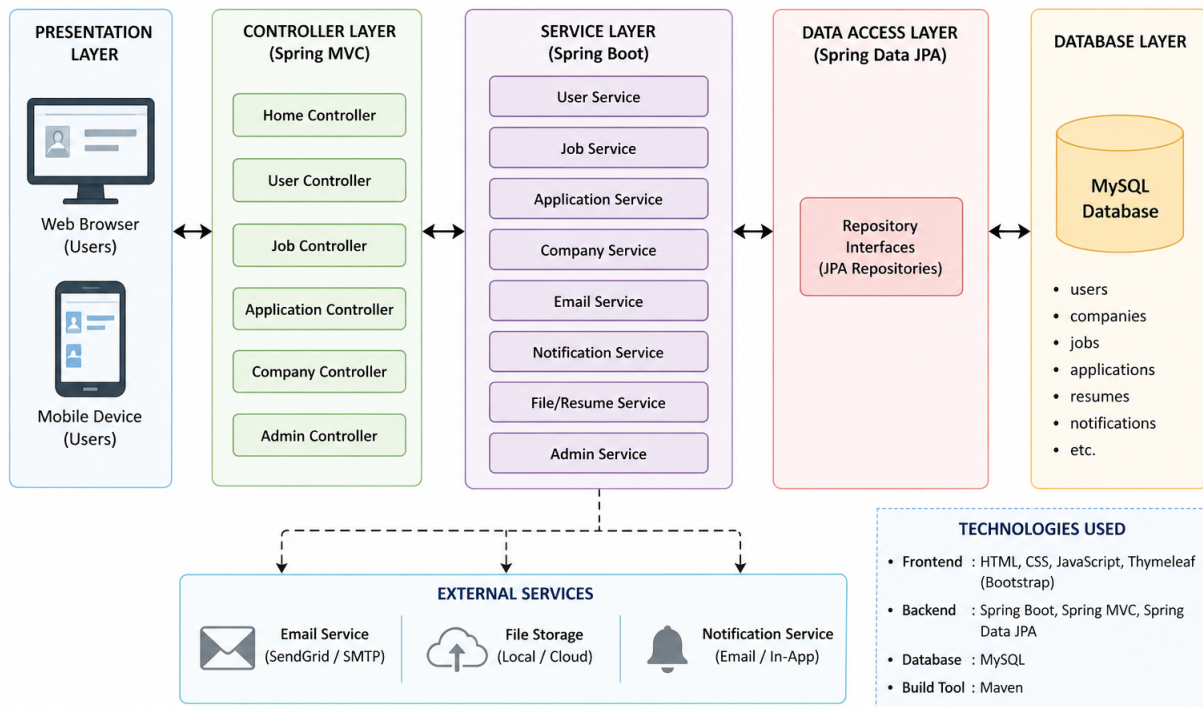


Figure 1.1: System Architecture of Job Portal Application

Figure 1.1 shows the system architecture of the Job Portal application, illustrating the layered structure of the system. The presentation layer consists of the user interface accessed through web browsers and mobile devices. The controller layer, implemented using Spring MVC, handles user requests and directs them to appropriate services. The service layer, built with Spring Boot, contains the business logic such as user management, job services, application handling, and notifications. The data access layer uses Spring Data JPA repositories to interact with the database efficiently. Finally, the database layer stores all relevant data including users, jobs, applications, and roles in MySQL. External services such as email notifications and file storage are also integrated to enhance system functionality. This architecture ensures scalability, modularity, and efficient data flow within the application.

Chapter 2

SURVEY OF SIMILAR APPLICATION IN MARKET

The rapid advancement of web technologies has led to the development of various online job portal platforms that connect job seekers with employers. These platforms provide efficient solutions for recruitment and job searching. A study of existing systems helps in understanding their features, advantages, and limitations, which in turn assists in designing an improved application.

Some of the popular job portal applications available in the market are:

2.1 LinkedIn

LinkedIn is one of the most widely used professional networking platforms. It allows users to create professional profiles, connect with industry experts, and apply for jobs. Recruiters can post job openings and search for candidates based on their skills and experience.

Advantages:

- Strong professional networking features
- Job recommendations based on user profiles

- Easy communication between recruiters and candidates

Limitations:

- Complex interface for new users
- Many features require premium subscription

2.2 Naukri

Naukri is a popular job portal widely used in India. It provides a platform for job seekers to upload resumes and apply for jobs, while employers can search for suitable candidates.

Advantages:

- Large database of job listings
- Resume upload and profile management
- Simple job search functionality

Limitations:

- Limited personalization of job recommendations
- Frequent notifications may be distracting

2.3 Indeed

Indeed is a global job search platform that aggregates job listings from various sources. It allows users to search for jobs based on location, salary, and job type.

Advantages:

- Wide range of job listings from multiple sources

- Easy-to-use interface
- Free access for job seekers

Limitations:

- Duplicate job postings may appear
- Limited direct interaction with employers

2.4 Comparison with Proposed System

The proposed Job Portal Application aims to overcome some of the limitations of existing systems by providing a simple, user-friendly interface and essential functionalities without unnecessary complexity. Unlike existing platforms, the system focuses on ease of use, efficient job management, and streamlined communication between users.

Key Improvements:

- Simplified user interface
- Focused features for academic-level implementation
- Efficient database management using MySQL
- Secure authentication using Spring Boot

Thus, the proposed system provides a balanced solution by combining essential features of existing applications while maintaining simplicity and efficiency.

Chapter 3

FRAMEWORK DESCRIPTION

3.1 Frontend Implementation

3.1.1 Environment Setup

The frontend of the application is developed using HTML, CSS, and Thymeleaf templates. Thymeleaf is a server-side Java template engine used for rendering dynamic web pages in Spring Boot applications.

The development environment includes:

- Visual Studio Code (VS Code) as the code editor
- Web browser (Chrome/Edge) for testing
- Maven for project dependency management

Thymeleaf integrates seamlessly with Spring Boot, enabling dynamic data rendering from the backend to the frontend.

3.1.2 Structure of the Project

The frontend files are organized within the project as follows:

- **templates/** – Contains Thymeleaf HTML files such as login, register, dashboard, and profile pages.

- **static/** – Contains static resources like CSS, JavaScript, and images.

Each HTML page is designed to interact with backend controllers and display dynamic content to users.

3.1.3 Execution Procedure

- Start the Spring Boot application.
- Open a web browser and navigate to the application URL (e.g., <http://localhost:8081>).
- The frontend pages are loaded dynamically using Thymeleaf.
- Users can interact with forms such as login, registration, and job applications.

3.2 Backend Implementation

3.2.1 Environment Setup

The backend is developed using Spring Boot, a Java-based framework that simplifies the creation of web applications.

The environment setup includes:

- Java Development Kit (JDK 17 or above)
- Spring Boot framework
- Maven for dependency management
- MySQL Driver (mysql-connector)

Spring Boot provides built-in support for REST APIs, dependency injection, and database connectivity.

3.2.2 Structure of the Project

The backend follows the MVC (Model-View-Controller) architecture and is organized into the following packages:

- **controller/** – Handles user requests and responses.
- **service/** – Contains business logic of the application.
- **repository/** – Handles database operations using JPA.
- **entity/** – Represents database tables as Java classes.
- **config/** – Contains configuration-related classes.
- **security/** – Manages authentication and authorization.

3.2.3 Execution Procedure

- Run the application using Maven command: `mvn spring-boot:run`.
- Spring Boot initializes the embedded server.
- Controllers handle incoming HTTP requests.
- Services process the business logic.
- Repositories interact with the database.
- Responses are sent back to the frontend.

The working flow of the system is illustrated in [Figure3.1]. It shows how requests move from the frontend to the backend and how responses are returned after processing.

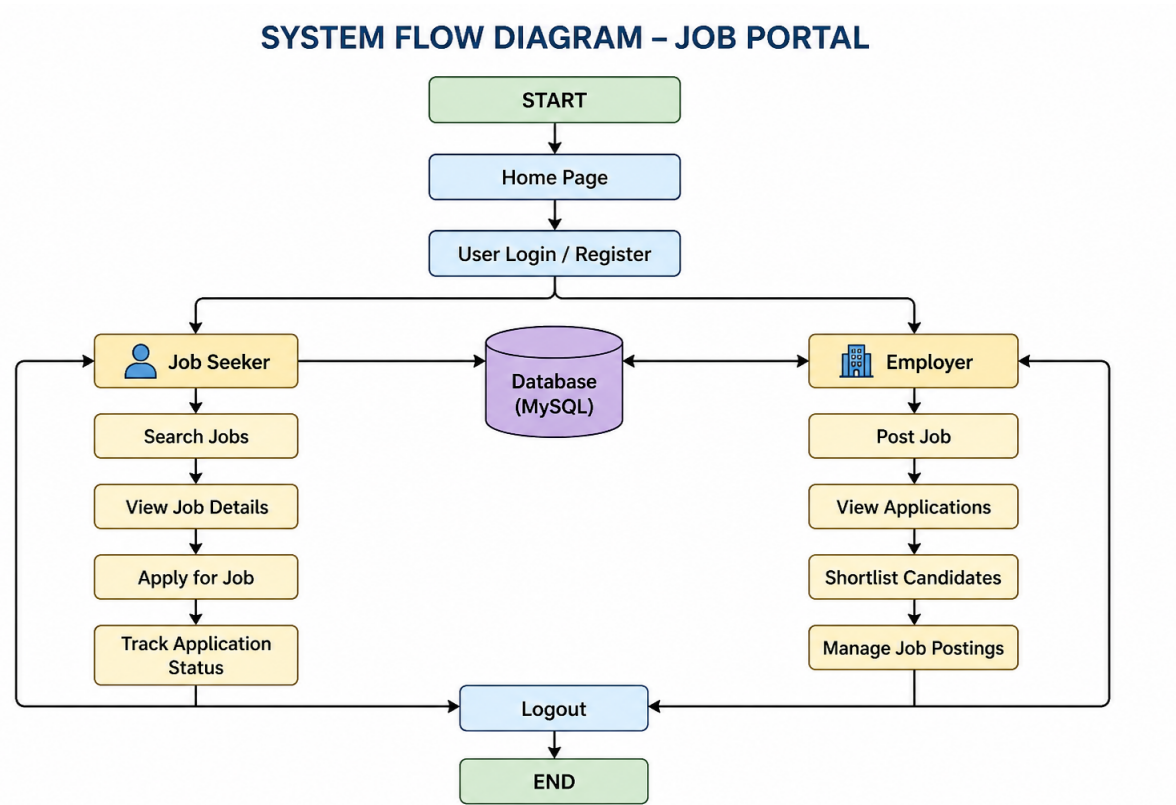


Figure 3.1: System Flow Diagram

Figure 3.1 shows the system flow diagram of the Job Portal, representing the overall working process of the application. The process begins with the home page, where users can either register or log in to the system. Based on the user role, the flow is divided into job seeker and employer functionalities. Job seekers can search for jobs, view job details, apply for jobs, and track their application status. Employers, on the other hand, can post jobs, view applications, shortlist candidates, and manage job postings. All the data is stored and retrieved from the MySQL database, ensuring efficient data management. The process concludes with the logout operation, marking the end of the session. This diagram clearly illustrates the interaction between different user roles and system components.

3.3 Database Implementation

3.3.1 Database Configuration

The database used in this project is MySQL. It is configured using the application.properties file in Spring Boot.

The configuration includes:

- Database URL (jdbc:mysql://localhost:3306/student_db)
- Username and password
- Hibernate properties for automatic table creation and updates

Spring Boot connects to MySQL using JDBC and manages data using JPA (Java Persistence API).

3.3.2 Schema Structure

The database schema is designed to store user and job-related information efficiently. It includes multiple tables that represent different entities in the system.

The schema ensures:

- Proper data organization
- Efficient querying
- Relationships between entities

3.3.3 Tables created

The main tables created in the database include:

- **users** – Stores user details such as ID, name, email, and password.
- **jobs** – Stores job postings including title, description, and company details.
- **applications** – Stores job application details linking users and jobs.

Chapter 4

METHODOLOGIES

4.1 Project Architecture

The project follows the Model-View-Controller (MVC) architecture, which separates the application into three main components: Model, View, and Controller. This separation improves maintainability, scalability, and code organization.

- **Model:** Represents the data layer, including entity classes and database interactions using JPA.
- **View:** Implemented using Thymeleaf templates to display dynamic content to users.
- **Controller:** Handles incoming requests, processes them using services, and returns appropriate responses.

The architecture ensures a smooth flow of data between the user interface and the database. The overall structure of the system is shown in [Figure4.1], which illustrates the interaction between different layers of the application.

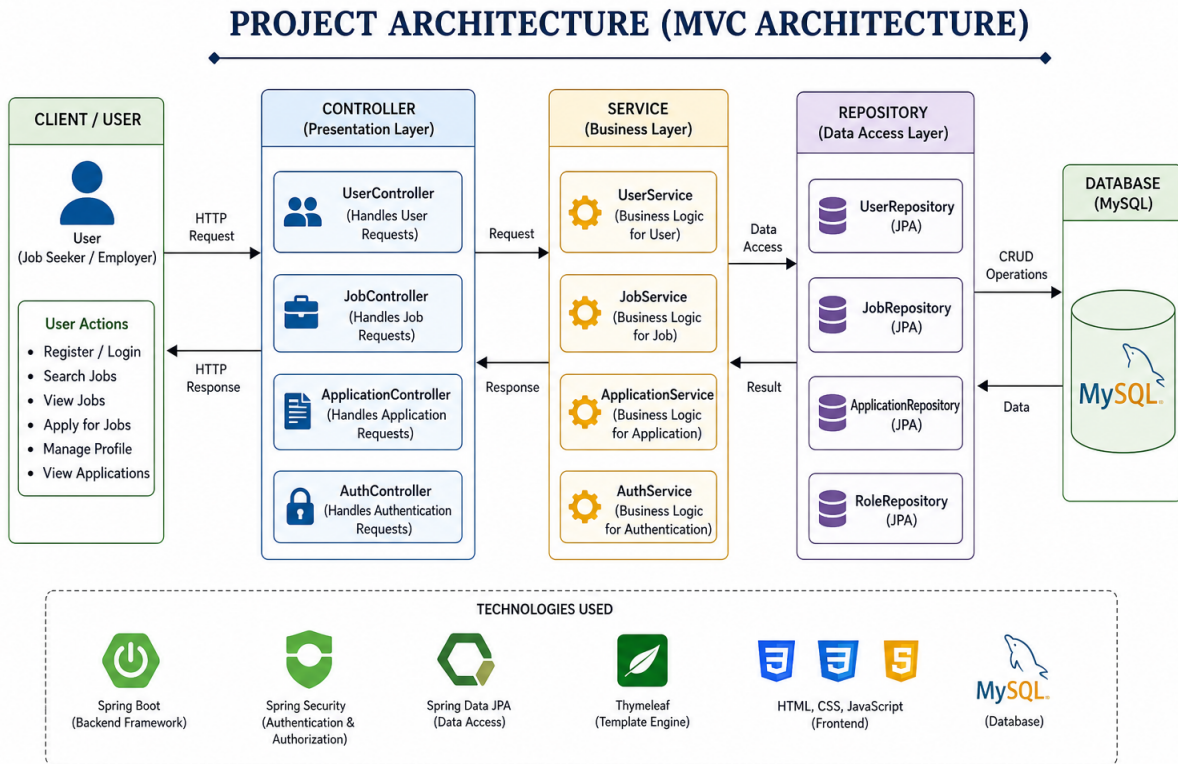


Figure 4.1: MVC Architecture of Job Portal System

Figure 4.1 shows The MVC architecture of the Job Portal system divides the application into three main components: Model, View, and Controller. The View layer, implemented using Thymeleaf templates, is responsible for presenting the user interface and displaying job-related information to users. The Controller layer, built with Spring Boot, handles user requests, processes business logic, and coordinates communication between the frontend and backend. The Model layer interacts with the database through JPA repositories, ensuring efficient storage and retrieval of job, user, and application data. This structured separation improves maintainability, scalability, and overall system performance.

4.2 DataFlow Diagram

The Data Flow Diagram (DFD) represents how data moves through the system. It shows the interaction between users, the application, and the database.

- Users provide input through the frontend (login, registration, job search).
- The controller processes the request and sends it to the service layer.
- The service layer interacts with the repository to access the database.
- The database returns the required data.
- The result is displayed to the user through the frontend.

The data movement within the system is illustrated in [Figure4.2], showing how information flows between different components during execution.

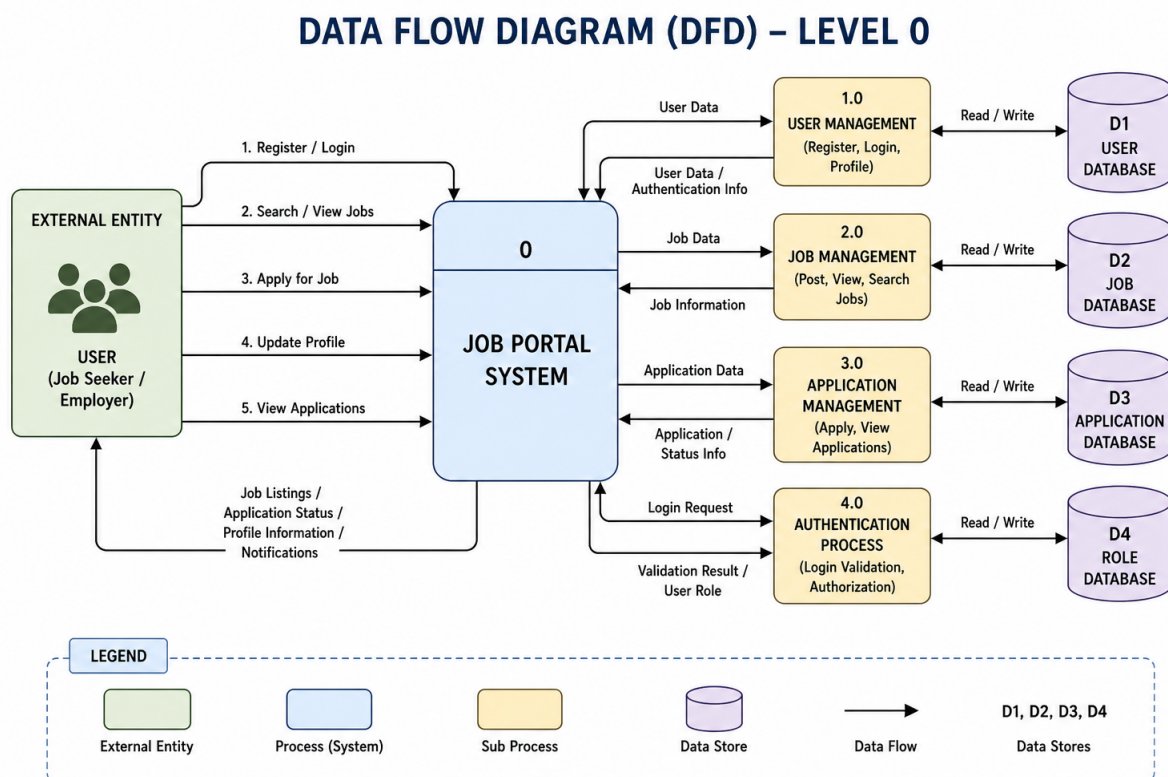


Figure 4.2: Data Flow Diagram

Figure 4.2 shows the Data Flow Diagram (DFD) Level 0 of the Job Portal System, illustrating the overall flow of data between users and the system. The external entity, which includes job seekers and employers, interacts with the system by performing actions such as registration, login, job search, job application, and profile management. The system processes these inputs through various modules including user management, job management, application management, and authentication. Each module communicates with corresponding data stores such as user database, job database, application database, and role database to store and retrieve relevant information. The diagram clearly represents how data flows between processes and storage components, ensuring efficient handling of user requests and maintaining data integrity within the system.

4.3 Module 1: User Management

This module handles user-related functionalities such as registration, login, and profile management.

- User registration with basic details
- Secure login authentication
- Profile viewing and updating

4.4 Module 2: Job Management

This module is responsible for managing job postings.

- Employers can post job vacancies
- Job details such as title, description, and company information are stored
- Users can browse available jobs

4.5 Module 3: Application Management

This module allows users to apply for jobs and manage applications.

- Users can apply for jobs
- Applications are stored in the database
- Employers can view applicants

Advantages of this Project

The proposed system provides several advantages:

- a) User-Friendly Interface:** The application provides a simple and easy-to-use interface for both job seekers and employers.
- b) Efficient Data Management:** All data is stored and managed using MySQL, ensuring quick access and reliability.
- c) Scalable Architecture:** The use of Spring Boot and MVC architecture allows the system to be easily extended with additional features.

Disadvantages

- Limited advanced features such as AI-based job recommendations.
- No email notification system for job updates.
- Requires internet connection for access.

Chapter 5

RESULTS AND DISCUSSIONS

The developed Job Portal Application was successfully implemented using Spring Boot, Thymeleaf, and MySQL. The system was tested for various functionalities including user registration, login, job posting, and job application[table 5.1].

The application provides a smooth and user-friendly interface, allowing users to easily interact with the system. The backend processes requests efficiently, and the database ensures secure and reliable data storage.

Observed Results:

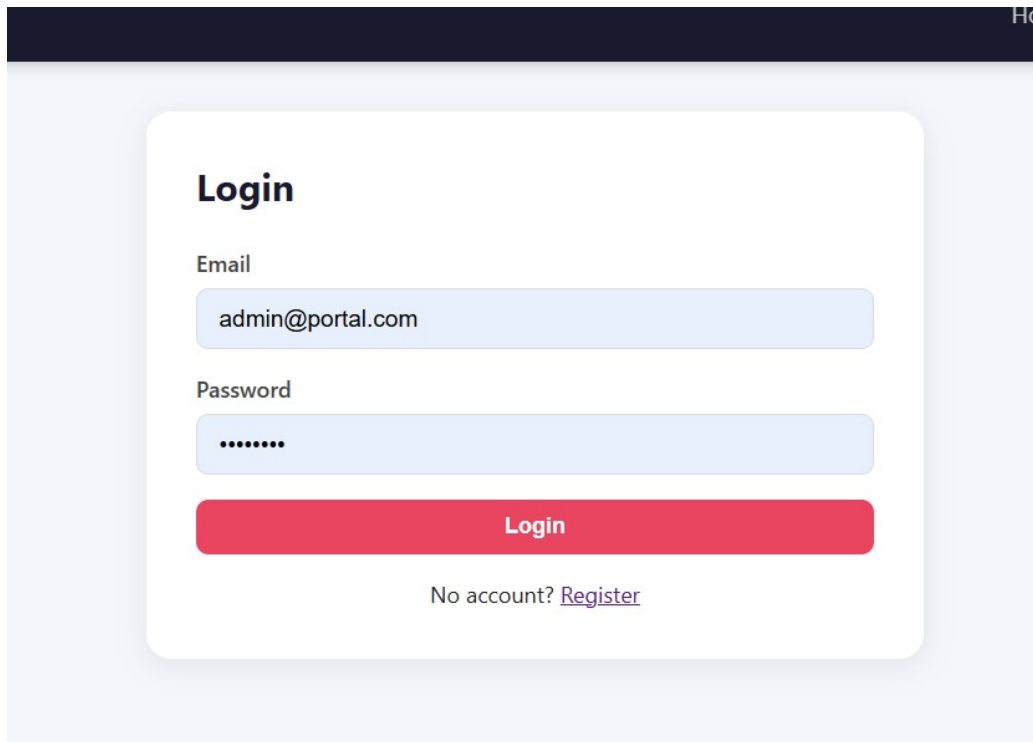
- Users are able to register and log in successfully.
- Employers can post job vacancies without errors.
- Job seekers can search and apply for jobs.
- Data is stored and retrieved correctly from MySQL database.
- The system performs efficiently with minimal response time.

Features	Existing Systems	Proposed System
User Interface	Complex	Simple and User-Friendly
Cost	Paid Features	Free to Use
Job Search	Available	Available
Job Application	Available	Available
Customization	Limited	Flexible
Performance	Moderate	Efficient

Table 5.1: Comparison of Existing Systems and Proposed System

Table 5.1 shows the comparison between existing systems and the proposed job portal system based on various features. The existing systems often have complex user interfaces and require paid features, whereas the proposed system provides a simple and user-friendly interface with free access. Both systems support job search and job application functionalities; however, the proposed system offers better customization and flexibility. Additionally, the performance of the proposed system is more efficient compared to the moderate performance of existing systems. This comparison highlights the improvements and advantages of the proposed system over traditional platforms.

5.1 System Screenshots and Results

A screenshot of a web application's login page. The page has a dark blue header bar at the top. Below it, a light blue rounded rectangle contains the login form. The form is titled "Login" in bold black text. It has two input fields: "Email" with the value "admin@portal.com" and "Password" with masked characters ".....". Below the password field is a red "Login" button. At the bottom of the form, there is a link that says "No account? [Register](#)".

Login

Email
admin@portal.com

Password
.....

Login

No account? [Register](#)

Figure 5.1: Login Output Screen

Figure 5.1 shows The login page allows users to securely access the job portal system. Users can enter their credentials to authenticate into the platform. The system validates inputs and prevents unauthorized access. This page acts as the entry point for all users.

The screenshot shows an 'All Applications' section with a table of 4 total applications. The table has columns for Applicant, Email, Job, Company, Applied, Status, and Action. Each row represents an application with details like applicant name, email, job title, company, application date, and status (pending). The Action column contains buttons for 'Accept', 'Reject', and 'Shortlist'.

Applicant	Email	Job	Company	Applied	Status	Action
raju	raju_123	data analyst	media tek	5/5/2026	pending	Accept Reject Shortlist
amar	ama_@_2123	Full Stack Developer	TechCorp Inc	20/4/2026	pending	Accept Reject Shortlist
amar	amar@gmail.com	Full Stack Developer	TechCorp Inc	20/4/2026	pending	Accept Reject Shortlist
sdv	asdfg	dish washer	dfgbnm	19/4/2026	pending	Accept Reject Shortlist

Figure 5.2: Employer Dashboard Output Screen

Figure 5.2 shows The employer dashboard displays job postings and applicant details. Employers can manage job listings efficiently from this interface. It allows adding, editing, and deleting job postings. This improves recruitment management and decision-making.

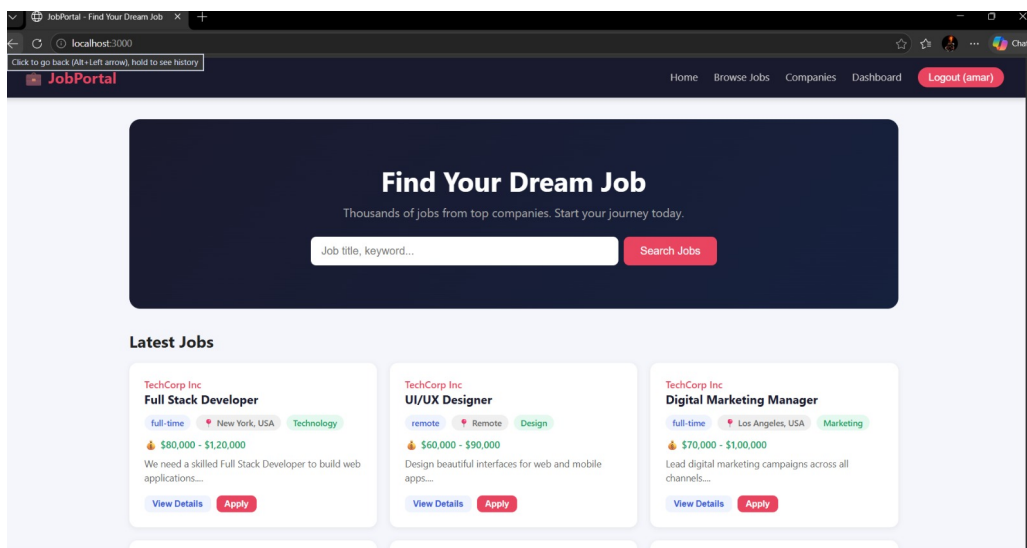
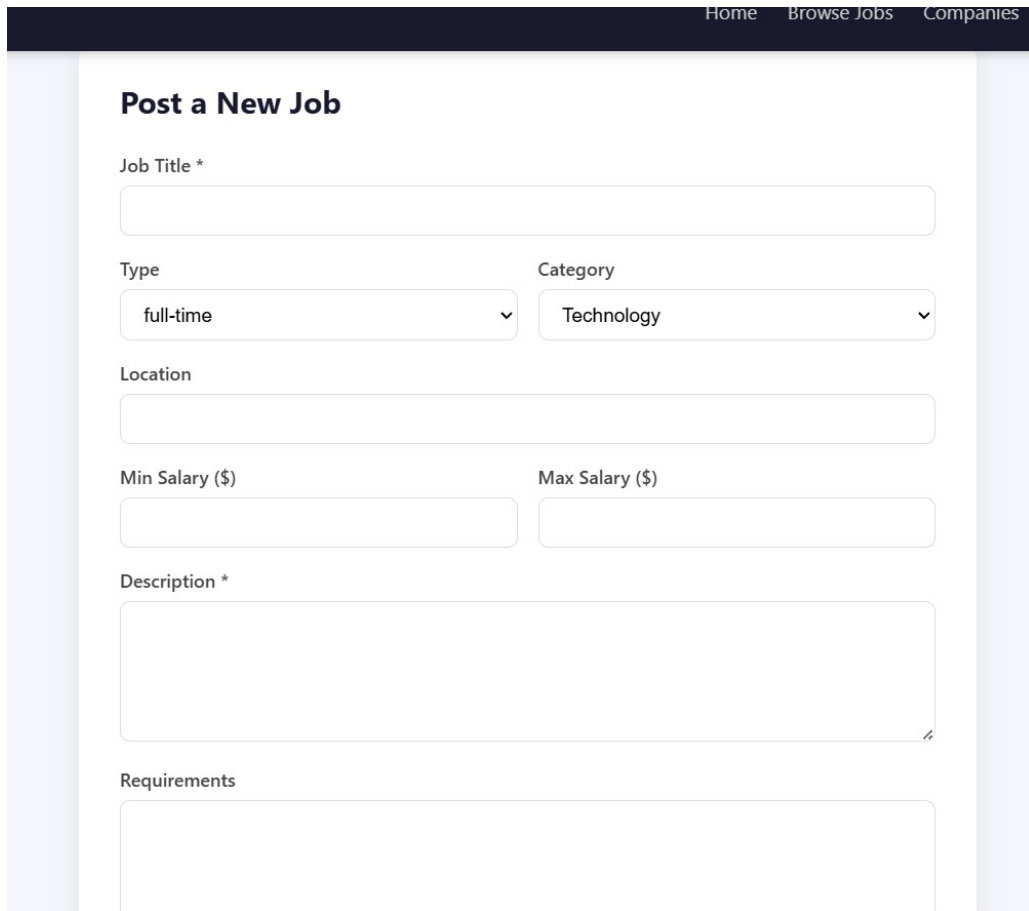


Figure 5.3: Edit Job Output Screen

Figure 5.3 shows The edit job page enables modification of job details. Users can update job title, description, and requirements. It ensures job listings remain accurate

and updated. This feature provides flexibility to administrators.



The screenshot shows a web application interface for posting a new job. At the top, a dark blue navigation bar contains the links 'Home', 'Browse Jobs', and 'Companies'. Below this, the main heading is 'Post a New Job'. The form consists of several input fields: a text field for 'Job Title *', a dropdown menu for 'Type' with 'full-time' selected, a dropdown menu for 'Category' with 'Technology' selected, a text field for 'Location', two text fields for 'Min Salary (\$)' and 'Max Salary (\$)', a large text area for 'Description *', and a text area for 'Requirements'. The form is set against a light blue background with a subtle vertical gradient.

Figure 5.4: Student Profile Output Screen

Figure 5.4 shows The student profile page stores user personal information. Users can upload resumes and update their skills. It helps recruiters evaluate candidates effectively. This ensures better job matching and selection.

Chapter 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 Conclusion

The Job Portal Application was successfully designed and implemented using Spring Boot, Thymeleaf, and MySQL. The system provides an efficient platform for job seekers and employers to interact, reducing the complexity of traditional recruitment processes.

The application supports essential functionalities such as user registration, login authentication, job posting, and job application. The use of MVC architecture ensures a well-structured and maintainable system. The integration of frontend and backend technologies enables smooth data flow and dynamic content rendering.

Overall, the project meets its objectives by providing a simple, scalable, and user-friendly solution. It demonstrates the effective use of modern web technologies in building real-world applications.

6.2 Future Enhancements

Although the system performs well, several improvements can be made in the future to enhance its functionality:

- **Email Notifications:** Notify users about job updates and application status.
- **Resume Upload Feature:** Allow users to upload and manage resumes.
- **Advanced Search Filters:** Improve job search with filters like salary, location, and experience.
- **Admin Dashboard:** Add admin control for managing users and job postings.
- **Mobile Application:** Develop an Android/iOS version of the system.
- **AI-based Recommendations:** Suggest jobs based on user skills and preferences.

6.3 Existing Applications and Review

- **LinkedIn**

<https://www.linkedin.com>

LinkedIn is a professional networking platform that allows users to build profiles, connect with professionals, and apply for jobs. It provides job recommendations based on user profiles and skills. However, it can be overwhelming for freshers due to its networking-heavy approach and competitive environment.

- **Naukri**

<https://www.naukri.com/>

Naukri is one of the most popular job portals in India, offering a wide range of job listings across various industries. It provides resume building, job alerts,

and recruiter connections. However, the platform often contains outdated job postings and requires premium services for better visibility.

- **Indeed**

`https://www.indeed.com/`

Indeed is a global job search platform that aggregates listings from multiple sources, making it easy for users to find opportunities. It offers company reviews, salary insights, and a simple application process. However, it lacks personalized filtering and may display duplicate or irrelevant job listings.

6.3.1 Application Access

The developed Event Booking System is currently hosted in a local environment (localhost) for testing and demonstration purposes. The application consists of the following modules, accessed in a sequential workflow:

- **User Registration Page**

URL: `http://localhost:8080/register`

Allows new users to create an account by entering their personal and login details.

- **User Login Page**

URL: `http://localhost:8080/login`

Enables registered users to securely log into the system using their credentials.

- **Job Portal Page**

URL: `http://localhost:8080/`

Displays available job listings and allows users to browse and apply for jobs.

- **User Application Page**

URL: `http://localhost:8080/jobs/2/apply`

Allows users to submit job applications by providing required details and resume.

- **Admin Dashboard**

URL: `http://localhost:8080/admin/dashboard`

Provides an overview of system activities including jobs, users, and applications.

- **Post New Job Page**

URL: `http://localhost:8080/jobs/new`

Allows administrators to create and publish new job postings.

- **Edit Job Page**

URL: `http://localhost:8080/jobs/1/edit`

Enables administrators to modify existing job details.

- **View Resume Page**

URL: `http://localhost:8080/admin/applications/4/resume`

Allows administrators to view resumes submitted by applicants.

Note: These URLs are accessible only within the local system environment and are used for development and testing purposes.

6.4 SOURCE CODE IMPLEMENTATION

Login Page

```
<!DOCTYPE html>
<html xmlns:th="http://www.thymeleaf.org">
<head>
<meta charset="UTF-8" />
<title>Login - Job Portal</title>
```

```

<link rel="stylesheet" th:href="@{/css/style.css}" />
</head>
<body>
<div class="container page-grid">
<header>
<div>
<h1>Job Portal </h1>
<p>Login to manage job applications , post roles , or browse jobs.</p>
</div>
<nav>
<a th:href="@{/}">Home</a>
<a th:href="@{/register}">Register </a>
</nav>
</header>

<section class="hero">
<div>
<h1>Welcome back </h1>
<p>Sign in to continue your journey.</p>
</div>
</section>

<section class="auth-form">
<h1>Sign In </h1>
<form method="post" th:action="@{/login}">
<label>Username </label>
<input type="text" name="username" required />

<label>Password </label>
<input type="password" name="password" required />

<button type="submit">Sign In </button>
</form>

<p>D o n t have an account?
<a th:href="@{/register}">Register </a></p>
<div th:if="${param.error}">Invalid username or password</div>
<div th:if="${param.logout}">Logged out successfully </div>
<div th:if="${param.registered}">Registration successful </div>

```

```
</section>
</div>
</body>
</html>
```

Student Dashboard

```
<!DOCTYPE html>
<html xmlns:th="http://www.thymeleaf.org">
<head>
<meta charset="UTF-8" />
<title>Student Dashboard</title>
<link rel="stylesheet" th:href="@{/css/style.css}" />
</head>
<body>

<div class="container page-grid">
<header>
<h1>Student Dashboard</h1>

<nav>
<a th:href="@{/}">Browse Jobs</a>
<a th:href="@{/applications}">My Applications</a>
<a th:href="@{/profile}">Profile</a>
<a th:href="@{/logout}">Logout</a>
</nav>
</header>

<section>
<h2>Available Jobs</h2>

<div th:each="job : ${jobs}">
<h3 th:text="${job.title}"></h3>
<p th:text="${job.description}"></p>

<p><strong>Location:</strong>
<span th:text="${job.location}"></span></p>
```

```

<p><strong>Employer:</strong>
<span th:text="${job.employer.fullName}"></span></p>

<form th:action="@{/jobs/' + ${job.id} + '/apply'}" method="post">
<button type="submit">Apply Now</button>
</form>

</div>
</section>

</div>
</body>
</html>

```

Application Tracking Page

```

<!DOCTYPE html>
<html xmlns:th="http://www.thymeleaf.org">
<head>
<meta charset="UTF-8" />
<title>My Applications</title>
<link rel="stylesheet" th:href="@{/css/style.css}" />
</head>
<body>

<div class="container page-grid">

<header>
<h1>My Applications</h1>

<nav>
<a th:href="@{/dashboard}">Dashboard</a>
<a th:href="@{/profile}">Profile</a>
<a th:href="@{/logout}">Logout</a>
</nav>
</header>

<section>

```

```

<div th:if="${applications.isEmpty()}">
No applications yet.
</div>

<div th:each="app: ${applications}">
<h3 th:text="${app.job.title}"></h3>

<p><strong>Employer:</strong>
<span th:text="${app.job.employer.fullName}"></span></p>

<p><strong>Status:</strong>
<span th:text="${app.status}"></span></p>

<p><strong>Applied On:</strong>
<span th:text="${app.appliedAt}"></span></p>

</div>

</section>
</div>

</body>
</html>

```

Chapter 7

ADDRESSING COMPLEX ENGINEERING PROBLEM AND SDG

7.1 Mapping of the Project to Complex Engineering Problem (CEP)

Level	Attribute	Characteristics	Wks	Justification
WP1	Depth of Knowledge	Requires in-depth engineering	WK3, WK4, WK5	Requires knowledge of Java, Spring Boot, JPA, and database design.
WP2	Conflicting Requirements	Technical and non-technical constraints	WK4, WK5	Balances usability and security.
WP3	Depth of Analysis	Analytical and design thinking	WK3, WK4	Includes ER design, APIs, resume upload.
WP4	Familiarity	Partially new problems	WK4	Job tracking beyond CRUD systems.
WP5	Applicable Codes	Limited standard solutions	WK6	Custom logic for resume and filtering.
WP6	Stakeholder Needs	Multiple stakeholders	WK5, WK6	Supports job seekers, recruiters, admin.
WP7	Interdependence	System integration	WK3, WK5	Integrates UI, backend, DB, validation.

Table 7.1: Mapping of CEP Attributes

7.2 Mapping of the Project to Complex Engineering Activities (CEA)

EA No.	Attribute	Characteristics	Justification
EA1	Range of Resources	Use of technologies	Java, Spring Boot, MySQL, HTML, CSS, JavaScript
EA2	Level of Interactions	Complex module interaction	Handles jobs, applications, login, and uploads
EA3	Innovation	Modern technologies	Role-based dashboards, filtering, email alerts
EA4	Consequences	Impact on users	Improves hiring and job accessibility
EA5	Familiarity	Beyond standard	Combines tracking, resume, and admin system

Table 7.2: Mapping of CEA Attributes

7.3 SDG Mapping

SDG No.	SDG Title	Relevance
SDG 8	Decent Work	Connects job seekers and employers for employment
SDG 9	Innovation	Uses modern web technologies for recruitment
SDG 10	Reduced Inequality	Equal job access for all users
SDG 16	Strong Institutions	Ensures security using authentication system

Table 7.3: SDG Mapping

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